

MOUNTAIN VILLAGE, AK, USA

WMO: 703056

Lat: 62.095N

Lon: 163.682W

Elev: 337

StdP: 14.52

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 26635

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.1	-17.4	-26.9	1.2	-18.6	-24.0	1.5	-15.3	39.1	13.9	35.7	18.4	7.0	330	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	72.3	59.2	67.6	57.2	63.9	55.4	60.6	69.3	58.4	65.5	56.4	62.3	10.4	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.1	70.4	62.2	55.1	65.5	60.4	53.5	61.7	58.0	27.0	69.8	25.4	65.1	24.2	62.5	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.9	28.5	25.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.5	5.9	10.6	12.6	23.4	39.6	50.3	54.3	52.3	45.1	32.1	16.4	10.4		
	DBStd	20.36	14.72	14.95	12.60	11.46	8.73	6.19	5.47	4.79	5.82	7.62	10.86	13.16		
	HDD50	7824	1368	1102	1159	798	337	69	15	28	158	555	1009	1226		
	HDD65	12951	1833	1522	1624	1248	788	442	335	393	596	1020	1459	1691		
	CDD50	353	0	0	0	0	14	78	149	100	12	0	0	0		
	CDD65	5	0	0	0	0	0	1	4	0	0	0	0	0		
	CDH74	57	0	0	0	0	3	12	41	1	0	0	0	0		
	CDH80	3	0	0	0	0	0	0	3	0	0	0	0	0		
	WSAvg	11.8	12.3	13.5	13.0	13.0	11.7	11.0	10.3	10.6	11.3	11.9	11.3	11.9		
	PrecAvg	24.9	1.7	1.8	1.5	1.3	1.3	1.7	2.4	3.2	3.1	2.6	2.5	2.0		
	PrecMax	33.9	3.6	4.9	4.8	2.6	2.8	2.9	4.4	6.2	6.1	4.3	5.7	4.2		
	PrecMin	19.9	0.4	0.5	0.3	0.3	0.4	0.6	0.9	1.2	0.2	0.5	0.3	0.7		
	PrecStd	3.5	0.9	1.1	1.0	0.6	0.6	0.4	0.8	1.0	1.1	0.9	1.4	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.3	36.0	37.5	51.8	71.1	75.5	79.2	72.8	61.6	50.2	38.9	39.5		
		MCWB	32.6	34.5	33.9	42.0	55.0	59.6	65.3	60.4	52.7	46.1	38.1	35.8		
	2%	DB	32.5	33.7	34.8	46.6	62.5	70.5	73.4	67.0	57.6	46.9	36.3	35.5		
		MCWB	31.7	32.7	32.9	39.3	50.3	57.4	60.4	57.3	50.3	44.5	35.4	33.8		
	5%	DB	30.3	32.2	32.3	41.6	57.2	65.6	68.8	63.9	55.1	44.7	34.1	32.3		
		MCWB	29.5	31.6	31.5	36.9	47.3	54.8	59.1	56.0	50.2	43.0	33.4	31.6		
	10%	DB	27.0	30.7	30.0	37.6	53.1	61.6	64.0	60.3	53.1	42.1	31.7	29.7		
		MCWB	26.3	30.2	29.6	34.3	45.5	52.6	56.6	54.6	49.7	40.8	31.1	29.1		
	0.4%	WB	33.3	34.8	34.9	42.4	56.3	61.8	66.4	61.5	54.6	47.2	38.5	37.4		
		MCDB	33.6	35.2	36.6	50.4	70.7	73.7	78.2	69.3	56.7	49.0	38.7	39.8		
	2%	WB	31.9	32.9	33.0	39.7	51.2	58.3	61.9	58.9	52.8	45.2	35.5	33.6		
		MCDB	32.5	33.3	34.4	46.0	61.0	67.4	70.5	65.3	55.3	46.6	35.9	35.1		
	5%	WB	29.7	31.8	31.7	37.0	48.3	55.7	59.3	57.0	51.1	43.2	33.5	31.6		
		MCDB	30.2	32.2	32.6	41.2	56.1	64.5	67.1	61.8	54.1	44.6	33.8	32.2		
	10%	WB	26.3	30.2	29.7	34.6	46.0	53.3	57.3	55.2	49.9	40.9	31.1	29.0		
		MCDB	27.2	30.7	30.1	37.3	52.3	60.0	63.3	59.2	52.7	42.2	31.7	29.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	11.4	12.2	12.9	15.0	15.1	12.6	11.5	10.5	8.2	8.6	9.8		
		MCDBR	10.3	8.7	9.7	16.3	22.5	22.5	22.0	17.2	13.4	9.1	7.5	9.5		
		MCWBR	10.2	8.3	8.6	10.5	12.6	11.5	11.3	9.1	8.4	7.1	7.2	8.6		
	5% WB	MCDBR	10.3	8.7	8.9	15.1	20.9	20.6	20.0	14.8	11.1	8.3	7.6	10.0		
		MCWBR	10.3	8.6	7.9	10.1	11.9	10.6	10.9	8.4	7.8	7.0	7.6	9.4		
(40) Clear-Sky Solar Irradiance	taub		0.209	0.269	0.276	0.347	0.370	0.362	0.367	0.380	0.327	0.288	0.231	0.197		
	taud		2.035	2.168	2.235	2.089	2.269	2.391	2.415	2.354	2.490	2.391	2.013	1.882		
	Ebn at Noon		154	220	267	263	268	273	269	253	250	215	134	86		
	Edh at Noon		11	21	28	40	37	34	32	31	23	17	11	7		
(44) All-Sky Solar Radiation	RadAvg		95	318	892	1417	1615	1664	1346	1071	705	344	110	44		
	RadStd		12	38	82	106	153	170	104	136	62	29	18	4		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	+3.01	+2.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page